

Zero-Knowledge Proofs

263-4665-00L

Lecturer: Jonathan Bootle

Lecturer

- Jonathan Bootle
- jonathan.bootle@inf.ethz.ch
- PhD on zero-knowledge at UCL
- Now at IBM Research – Zurich
- Started studying cryptography to make a secure secret-santa protocol!



Teaching assistant

- David Balbás
- david.balbasgutierrez@inf.ethz.ch
- Postdoc in the ETH Applied Crypto group.
- PhD in pairing-based cryptography.
- Can prove in zero-knowledge that the Spanish olive oil is the best one in the world.



Teaching assistant

- Stan Castellana
- scastellana@student.ethz.ch
- BSc at EPFL and University of Edinburgh
- MSc at ETH
- Driven by cryptography's ability to reshape power dynamics, with a focus on building systems that protect digital autonomy and privacy.



Time and Location

- Lectures on Fridays, 12:00-14:00, CHN G 42.
- Exercise sessions on Fridays, 15:00-16:00, in CHN F 42.
- Online on Zoom, and recorded.
- **Exception:** today's exercises/demonstration
- Office hours poll closes 23:59, 21/09

 Ankündigungen

 Questions and Discussions

 Office Hours Poll

 Hidden from students

Opens: Friday, 18 September 2026, 12:00 PM

Closes: Monday, 21 September 2026, 11:59 PM

 Zoom Link for Lectures (Fri 12-14:00)
Starts: Friday, 18 September 2026, 12:00 PM

 Zoom Link for Exercise Sessions (Fri 15-16:00)
Starts: Friday, 25 September 2026, 3:00 PM

Materials

- Posted on Moodle
- Lecture slides, shortly before each lecture
- Exercise sheets, about a week before each exercise session
- Lecture and exercise session recordings, once processed
- Links to external resources such as papers and notes

(from a previous year)

▼ 20/09 Week 1

 Lecture 1 Video

 Lecture 1 Slides

 Exercises 1 Slides

 Resources W01

(from a previous year)

Examination

▼ 17/10 Submission for Exercise 5 (graded)

 Graded Exercises for Week 5

 Submission for Exercise 5 (graded)

- This course = 5 ECTS credits
- 70% session examination, 2 hour written exam
 - Practice exam questions available later in the course
- 30% graded exercise sheets
 - 10% week 5 – submit through Moodle on or before Thursday 15/10/2025
 - 10% week 9 – submit through Moodle or before Thursday 12/11/2025
 - 10% week 13 – submit through Moodle on or before Thursday 10/12/2025
- Other weeks ungraded, but **essential** for practice and final exam

Goals of the course

- To understand what it means for a zero-knowledge proof to be secure
- To construct and analyse various types of zero-knowledge proofs
- To understand some applications of zero-knowledge proofs

Prerequisites

- Familiarity with cryptography (e.g. security games, p.p.t. adversaries, indistinguishability as in IND-CPA security)
- Groups, finite fields, modular arithmetic
- Vectors and matrices
- **Polynomial arithmetic over e.g. $\mathbb{F}[X]$, $\mathbb{F}[X_1, X_2, X_3]$**
- Probability
- May use all of the above **at the same time**
- Proofs: ~~deep difficult~~ messy
- Review maths at <https://shoup.net/ntb/ntb-v2.pdf>

Why you should study zero-knowledge proofs

- Philosophy: new perspectives on proof, knowledge and learning
- Theory: simulation techniques, links to complexity theory
- Practice: used in digital signatures, e-voting, mix nets, verifiable computation, verifiable machine learning, blockchains and more...
- Because they are interesting!
- If you are ready for a challenge...

Course Outline (14 lectures)

1. Introduction and definitions ~2 lectures

2. Sigma protocols ~3 lectures

3. ZK arguments with short proofs ~5 lectures

4. Non-interactive zero-knowledge ~3 lectures

5. Bonus material? ~1 lecture

Lecture 1: Interactive proofs and zero-knowledge

What happens when you add interaction, randomness, and cryptographic assumptions to mathematical proofs?

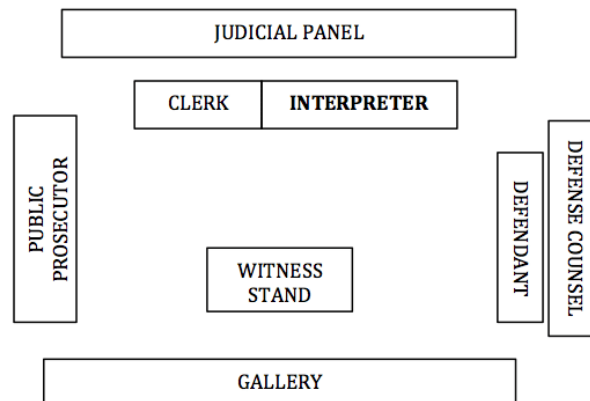
Types of proof

Proof:

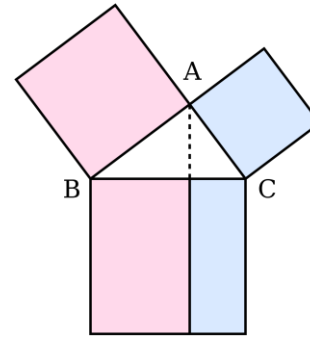
“the process or an instance of establishing the validity of a statement especially by derivation from other statements in accordance with principles of reasoning”

<https://www.merriam-webster.com/dictionary/proof>

Real life



Formal



This course

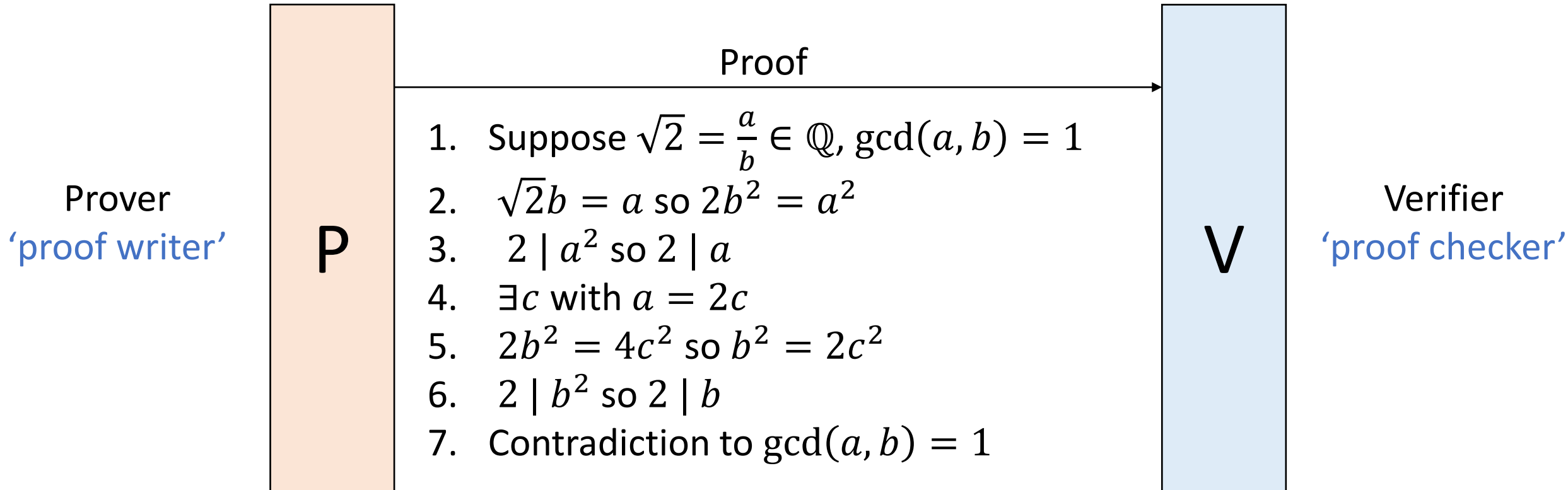
Static
Deterministic

Interactive
Randomised

Known for over
2000 years!

Mathematical proofs

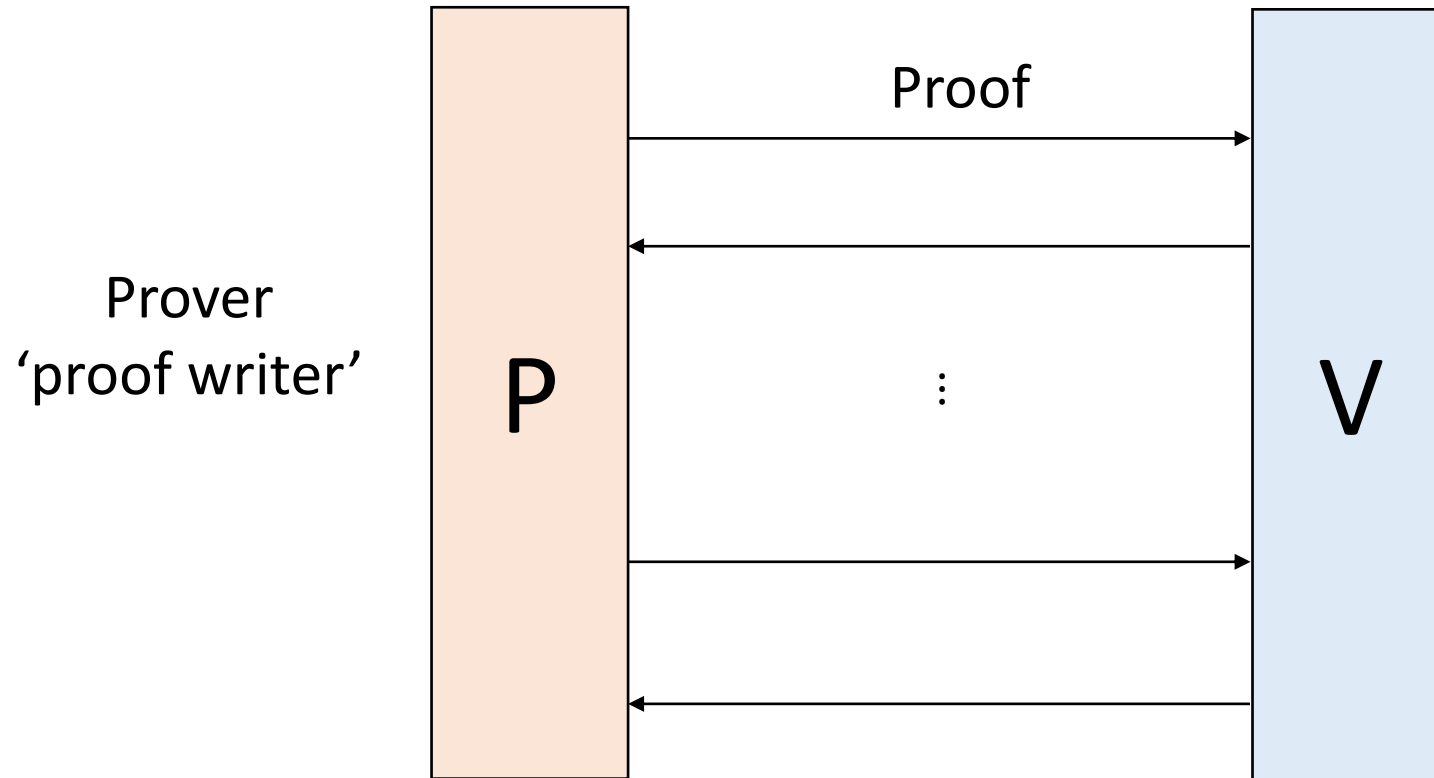
Statement to prove $\sqrt{2} \notin \mathbb{Q}$



- All errors detectable by V
- The same 1-directional message every time
- Sequence of logical assertions
- Each must be implied by earlier ones

Interactive proofs (IP)

Statement to prove



- Most errors detectable by V
- Randomised, adaptive, 2-directional messages

“The Knowledge Complexity of Interactive Proof Systems”, 1985



Verifier
'proof checker'

1. Increased proving power
2. Counterintuitive properties
3. Drastic efficiency gains

1,2 today, 3 in Sections 3 and 4

OpenAI says it cracked 90-year-old maths problem in 88 hours

8 September 2026

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The ChatGPT-maker said on Tuesday that by focusing a group of roughly 10,000 AI agents, or AI bots that undertake tasks somewhat autonomously, it solved a notoriously difficult maths problem in just 88 hours.

Tristan Buckmaster, a mathematics professor at New York University, said on Tuesday that he and Levent Alpöge, a mathematician working for OpenAI rival Anthropic, had also been working on solving the problem.

The duo had been using OpenAI's tool Codex in their work. But Buckmaster said that on 3 September, he found out that "information about our progress had been passed to OpenAI".

"While unlikely, we cannot rule out that de-identified data derived from their usage of our products helped improve our models", the company added. "However, our proofs differ significantly and even the precise results proved are different."

Prove that a model was not trained on specific data!

2. Counterintuitive properties – no information about model or actual training data need be leaked
3. Drastic efficiency gains – verification exponentially cheaper than training

<https://www.bbc.com/news/articles/cy7zygy3rl2o>

Agenda

- Definitions of complexity classes and IPs
- IP for graph non-isomorphism – increased proving power
- IP for graph isomorphism – counterintuitive properties

Complexity classes

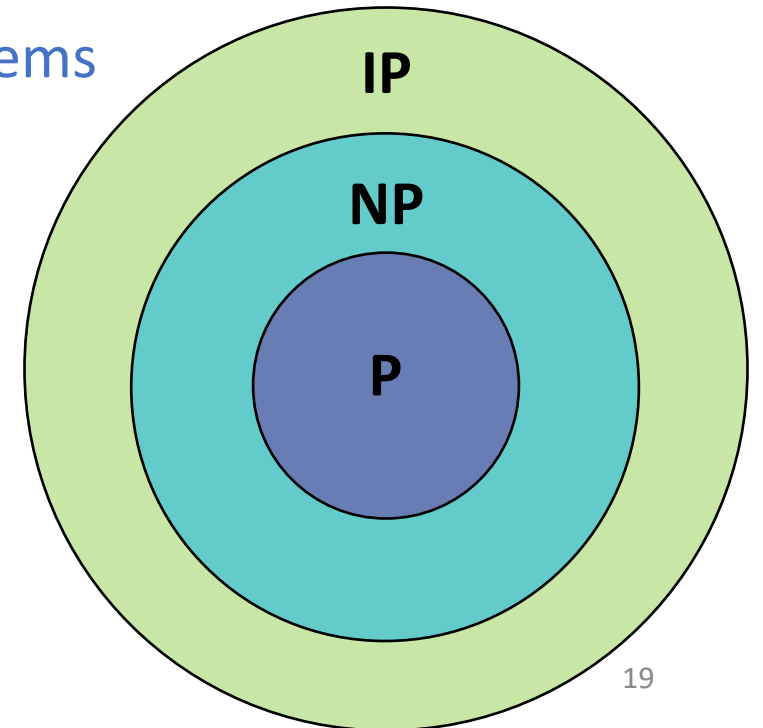
increased proving power = proofs for larger classes of statements
complexity classes

Definition:

- A *statement/instance* is an element $x \in \{0,1\}^*$. binary encoding of a YES/NO problem statement
- A *language* is a set $\mathcal{L} \subseteq \{0,1\}^*$. collection of problems
- A *complexity class* is a set of languages.
problem types with similar properties

Example:

- $\mathcal{L}_{G3C} = \{\text{graphs } G \text{ which have a 3-colouring}\}$.
Problem: decide whether G has a 3-colouring or not



The complexity class $\mathbf{P}$

Definition:

$\mathbf{P}$ is the set of languages $\mathcal{L}$ for which

$\exists$ decision algorithm M and polynomial q

i.e. Turing Machine

such that $\forall x \in \{0,1\}^*$,

- $M(x) = 1 \iff x \in \mathcal{L}$; and
- M finishes in $\leq q(|x|)$ steps on input x . i.e. polynomial time, efficient

Problems which are
easy to solve



Efficient checks
that $x \in \mathcal{L}$

The complexity class NP

Definition:

NP is the set of languages $\mathcal{L}$ for which

$\exists$ language $\mathcal{R}_{\mathcal{L}}$ and polynomial q such that

- $\forall x \in \mathcal{L}, \exists w$ with $|w| \leq q(|x|)$ and $(x, w) \in \mathcal{R}_{\mathcal{L}}$;
- $\forall x \notin \mathcal{L}, \nexists w$ with $(x, w) \in \mathcal{R}_{\mathcal{L}}$; and
- $\mathcal{R}_{\mathcal{L}} \in \mathbf{P}$.

We call w the *witness* to x .
a solution to x

Problems whose YES
solutions are easy to check

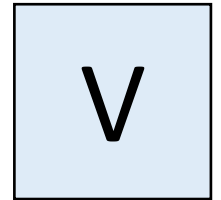
Efficient checks
that $(x, w) \in \mathcal{R}_{\mathcal{L}}$

It could be hard to find w (if $\mathbf{P} \neq \mathbf{NP}$).

Traditional proofs and complexity classes

Task: prove that $x \in \mathcal{L}$.

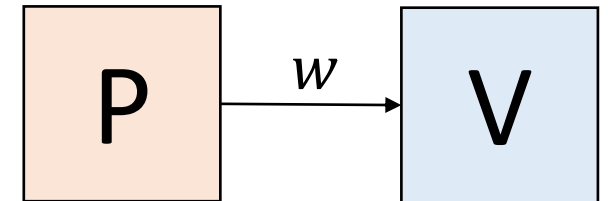
$\mathcal{L} \in \mathbf{P} \iff x \in \mathcal{L}$ is obvious and needs no proof!



Efficient checks that $x \in \mathcal{L}$

$\mathcal{L} \in \mathbf{NP} \iff x \in \mathcal{L}$ has a proof which is easy to check

w may be hard to find



Efficient checks that $(x, w) \in \mathcal{R}_{\mathcal{L}}$
 $x \notin \mathcal{L} \Rightarrow (x, w) \notin \mathcal{R}_{\mathcal{L}}$

2.1 Interactive Turing machines and interactive pairs of Turing machines

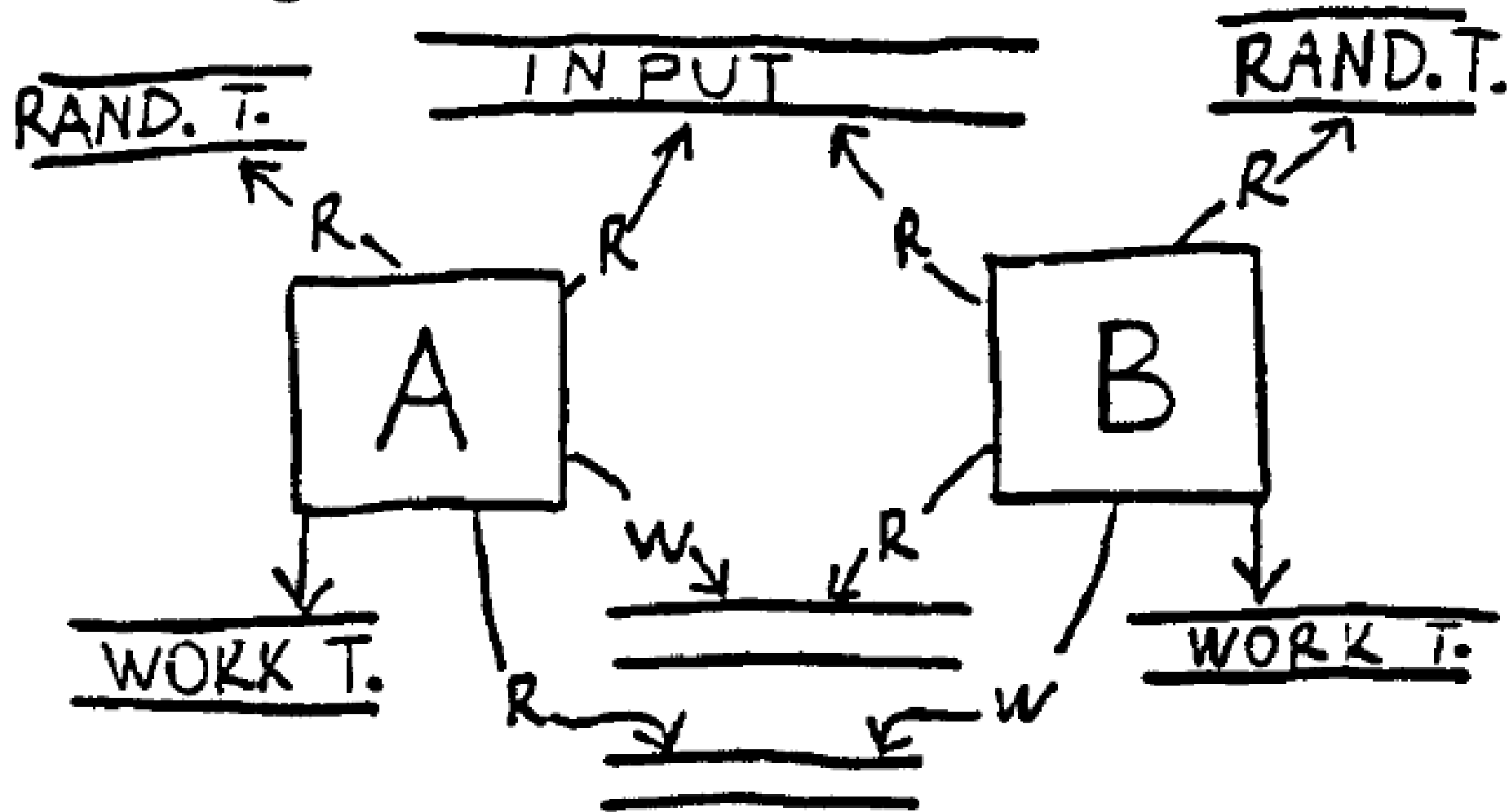


Fig. 2: an interactive pair of Turing machines

Interactive algorithms

Definition: Functions, Turing machines, circuits

Let $P, V : \{0,1\}^* \rightarrow \{0,1\}^*$. Let $k : \{0,1\}^* \rightarrow \mathbb{N}$.

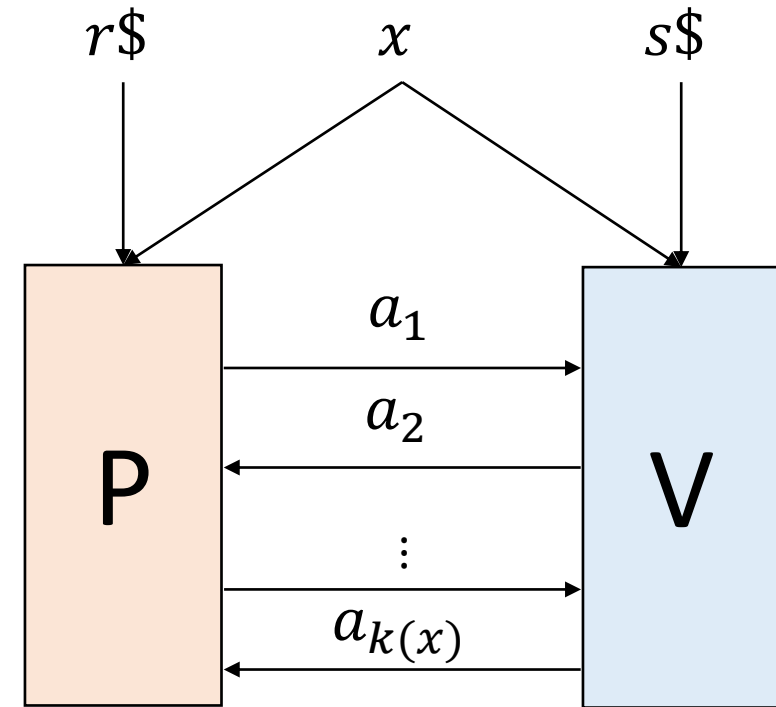
A $k(x)$ -move interaction between P and V

on input $x \in \{0,1\}^*$ is defined by:

- $a_1, st_{P,1} = P(x, r)$
- $a_2, st_{V,2} = V(x, a_1, s)$
- $a_3, st_{P,3} = P(x, st_{P,1}, a_2, r)$
- $\vdots$
- $a_{k(x)}, st_{V,k(x)} = V(x, st_{V,k(x)-2}, a_{k(x)-1}, s)$

The interaction is denoted $\langle P(r), V(s) \rangle(x) = a_{k(x)}$.

The *transcript* is $(x, a_1, \dots, a_{k(x)})$.



Any constants with a gap define the same complexity class

The complexity class $\mathbf{IP}$

Same complexity class even if P doesn't have r (later exercise sheet)

P need not be efficient!

Total running time of V

Definition:

$\mathcal{L} \in \mathbf{IP}$ if $\exists P$ and V with V efficient (polynomial time in $|x|$) satisfying:

- *Completeness*

$$\forall x \in \mathcal{L}, \Pr_{r,s}[\langle P(r), V(s) \rangle(x) = 1] \geq 3/4$$

“True statements usually accepted”

- *Soundness*

$$\forall x \notin \mathcal{L}, \forall P^*, \Pr_{r,s}[\langle P^*(r), V(s) \rangle(x) = 1] \leq 1/2$$

“False statements usually rejected”

If so, we say that (P, V) is an *interactive proof system* for $\mathcal{L}$.

Agenda

- Definitions of complexity classes and IPs ✓
- **IP for graph non-isomorphism – increased proving power**
- IP for graph isomorphism – counterintuitive properties

Graph Non-Isomorphism Problem

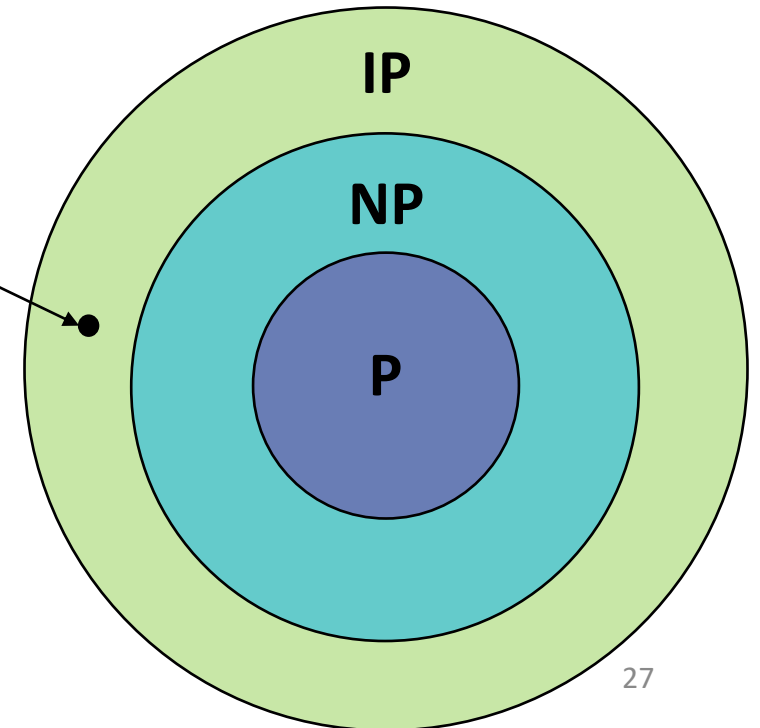
Definitions:

- Two graphs G_0 and G_1 on vertex set $[n]$ are *isomorphic* (written $G_0 \cong G_1$) if there exists a permutation $\pi : [n] \rightarrow [n]$ such that $G_0 = \pi(G_1)$.

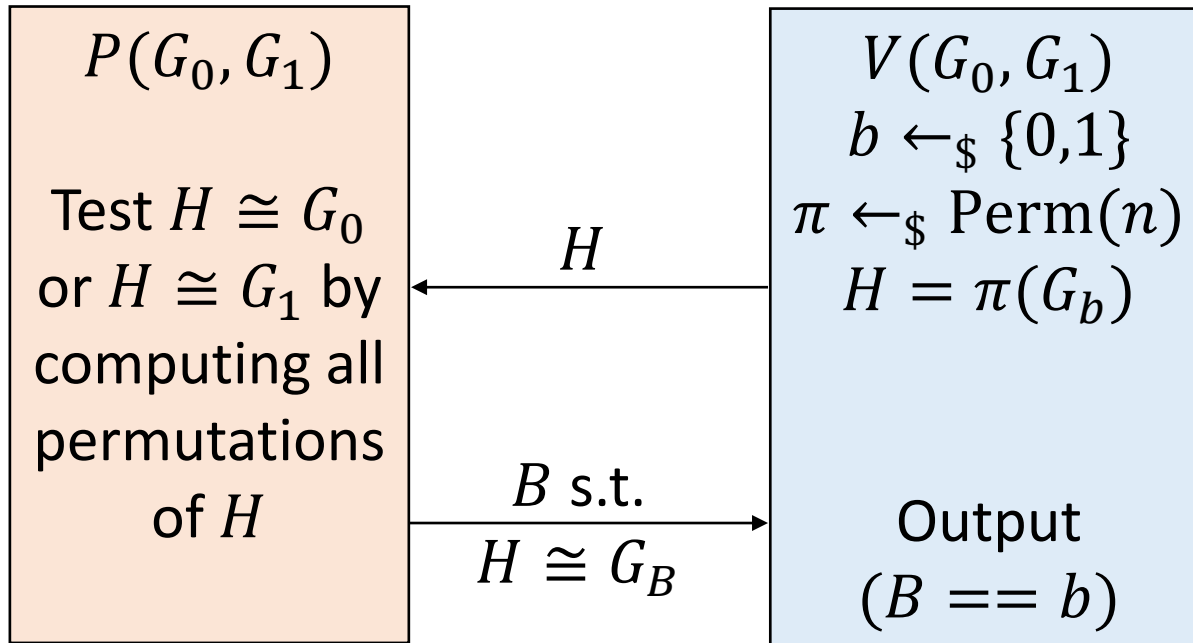
π relabels vertices

- $$\mathcal{L}_{GNI} := \left\{ (G_0, G_1) : \begin{array}{l} G_0, G_1, \text{ vertex set } [n] \\ G_0 \not\cong G_1 \end{array} \right\}$$

$\exists$ an interactive proof for $\mathcal{L}_{GNI}$
 $\mathcal{L}_{GNI} \notin \mathbf{NP}$ (as far as we know)
Evidence that $\mathbf{NP} \neq \mathbf{IP}$.



Interactive proof for $\mathcal{L}_{GNI}$



Security analysis:
Traditional mathematical
proofs of completeness
and soundness

Completeness: $G_0 \not\cong G_1$

- $(H \cong G_0) \wedge (H \cong G_1) \Rightarrow G_0 \cong G_1 \#$
- So $H \cong G_b$ and $H \not\cong G_{1-b}$
- P must choose $B = b$ so V accepts

Always accepts

Soundness: $G_0 \cong G_1$

- $\pi(G_0), \pi(G_1)$ identically distributed
- B, b independent
- $\Pr[B = b] = 1/2 \leq 1/2$

Agenda

- Definitions of complexity classes and IPs ✓
- IP for graph non-isomorphism – increased proving power ✓
- **IP for graph isomorphism – counterintuitive properties**

Transfer of knowledge

$$\sqrt{2} \notin \mathbb{Q}$$

1. Suppose $\sqrt{2} = \frac{a}{b} \in \mathbb{Q}$, $\gcd(a, b) = 1$
2. $\sqrt{2}b = a$ so $2b^2 = a^2$
3. $2 \mid a^2$ so $2 \mid a$
4. $\exists c$ with $a = 2c$
5. $2b^2 = 4c^2$ so $b^2 = 2c^2$
6. $2 \mid b^2$ so $2 \mid b$
7. Contradiction to $\gcd(a, b) = 1$

Hard to produce if you
didn't know it already

knowledge
gained

Now easy to produce and generalise

Transfers knowledge from P to V

You can now generalise to $\sqrt{3}, \sqrt{5}, \dots$

Can we avoid this?

Easy to produce if you
did know it already

no knowledge
gained

Working definition of algorithmic knowledge

Informal Definition:

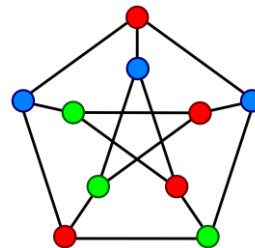
IP definition: P unbounded, V efficient

An algorithm's *knowledge* is anything that it can compute efficiently.

Examples:

1. Your birthday
2. 100th decimal digit of $\frac{\text{your birth month}}{\text{your birth year}} \cdot \sqrt{2}$
3. 43rd digit of π
4. 3-colouring of a large graph

NP-complete



Every-day	Algorithmic
Yes	Yes
No	Yes
Not usually	Yes
No	Probably not

Zero-knowledge

No knowledge gained by verifiers who could already produce the proof by themselves.

V may not follow the protocol

Definition:

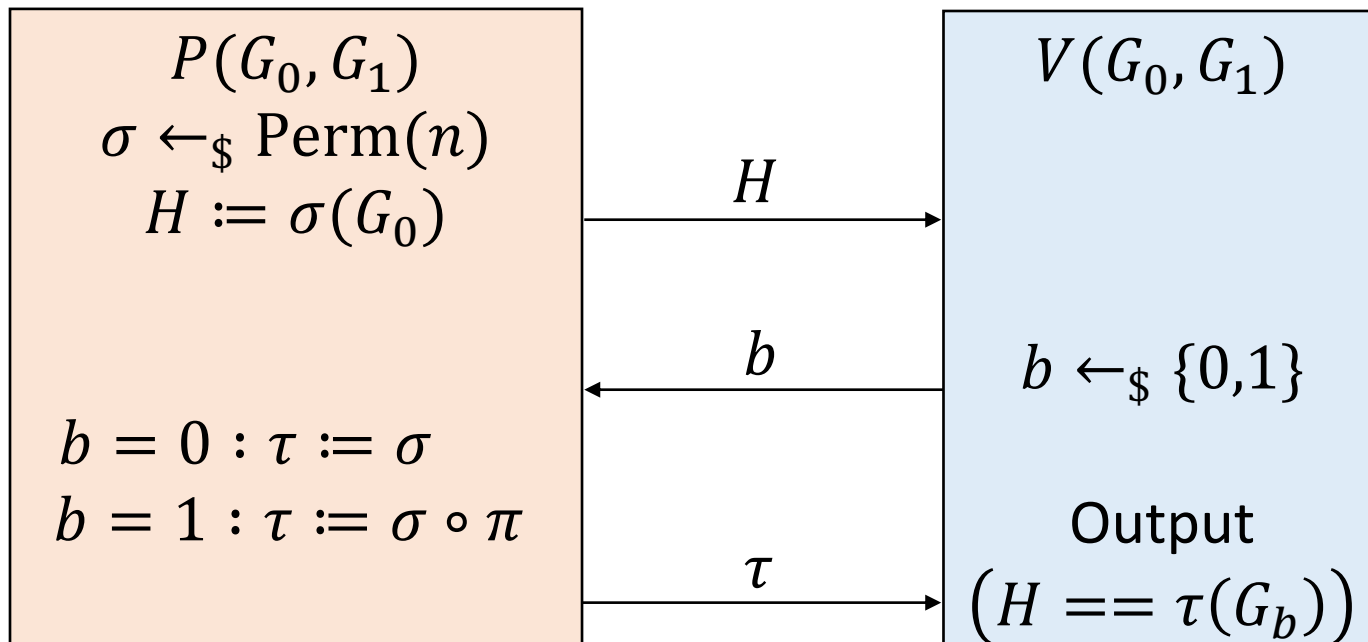
Let (P, V) be an IP for $\mathcal{L}$.

- The *verifier's view* is $\text{View}_V^P = (x, s, a_1, \dots, a_{k(x)})$. everything V sees
- (P, V) is *perfect zero-knowledge* if $\forall$ efficient V^* , $\exists$ efficient simulator S such that $\forall x \in \mathcal{L}$, we have $\{\text{View}_{V^*}^P\} = \{S(V^*, x)\}$. expected probabilistic polynomial time
- If so, (P, V) is a *perfect zero-knowledge proof (perfect ZKP)*. equal as probability distributions

Note: $\text{View}_{V^*}^P$ of malicious verifier V^*

Perfect ZKP for Graph Isomorphism

- $\mathcal{L}_{GI} := \{(G_0, G_1) : G_0, G_1 \text{ graphs}, G_0 \cong G_1\}$.
- $\mathcal{R}_{GI} := \{((G_0, G_1), \pi) : (G_0, G_1) \in \mathcal{L}_{GI}, G_0 = \pi(G_1)\}$. Efficiently checkable
so $\mathcal{L}_{GI} \in \mathbf{NP}$



Not known to be **NP**-complete

Almost in **P** (!)

Prove completeness,
soundness and perfect ZK

Domain checks e.g. $\tau \in \text{Perm}(n)$
sometimes extremely important
but usually omitted

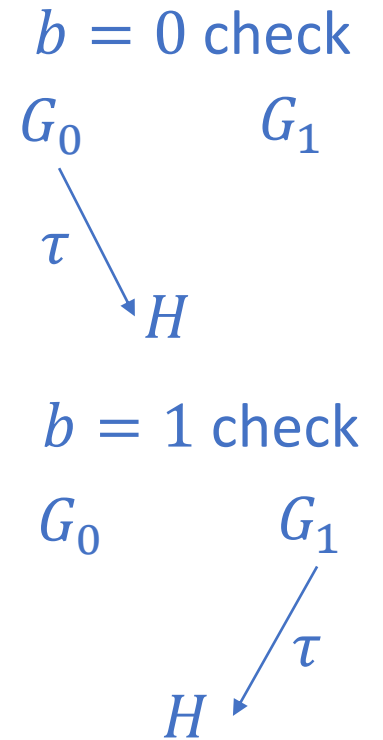
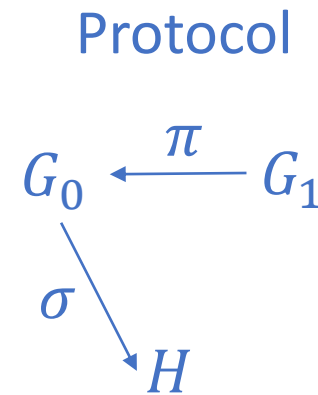
Completeness and soundness analysis

Completeness: $x \in \mathcal{L}_{GI}, G_0 \cong G_1$

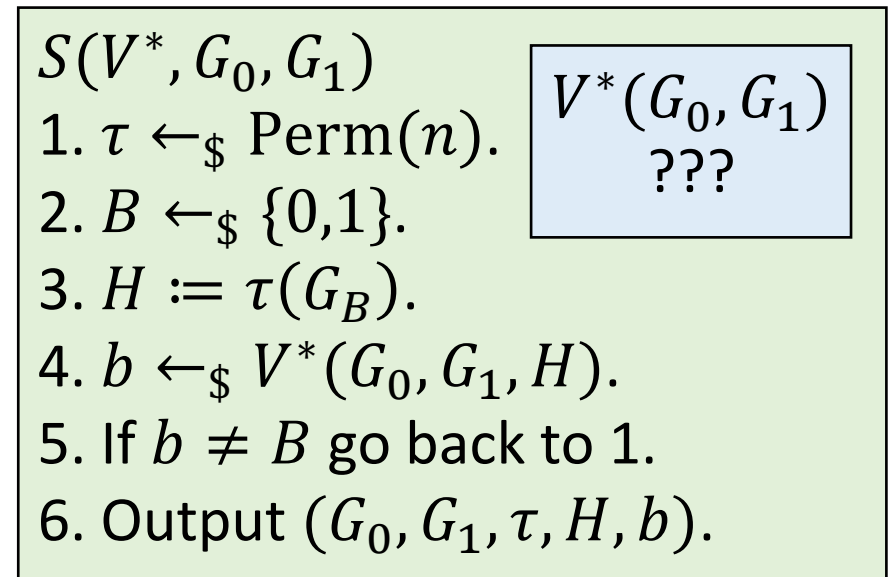
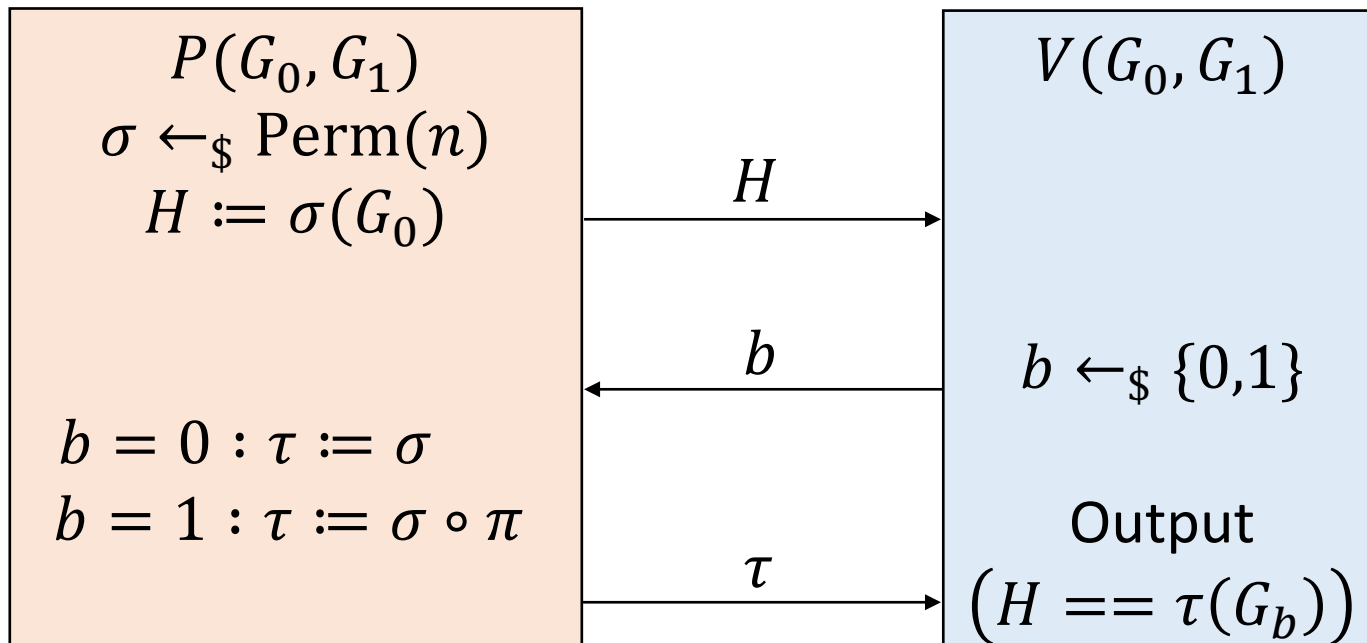
- If $b = 0$ then $\tau = \sigma$.
- If $b = 1$ then $\tau = \sigma \circ \pi$.
- So V always accepts.

Soundness: $x \notin \mathcal{L}_{GI}, G_0 \not\cong G_1$

- No graph H has both $G_0 \cong H$ and $G_1 \cong H$.
- Otherwise $G_0 \cong G_1$ #
- After P sends H , we have $G_b \cong H$ for at most one of $b = 0, 1$.
- With probability $\geq 1/2$, V chooses b with no isomorphism and outputs 0



Simulation of the protocol



S works backwards
and guesses b

Perfect zero-knowledge analysis

What does the verifier see?

- Real protocol: $View_{V^*}^P = (G_0, G_1, \tau, H, b)$.
- $\tau = \sigma$ or $\tau = \sigma \circ \pi$ is uniform from $\text{Perm}(n)$.
- τ, H, b random satisfying $H = \tau(G_b)$.

$S(V^*, G_0, G_1)$

1. $\tau \leftarrow_{\$} \text{Perm}(n)$.
2. $B \leftarrow_{\$} \{0,1\}$.
3. $H := \tau(G_B)$.
4. $b \leftarrow_{\$} V^*(G_0, G_1, H)$.
5. If $b \neq B$ go back to 1.
6. Output (G_0, G_1, τ, H, b) .

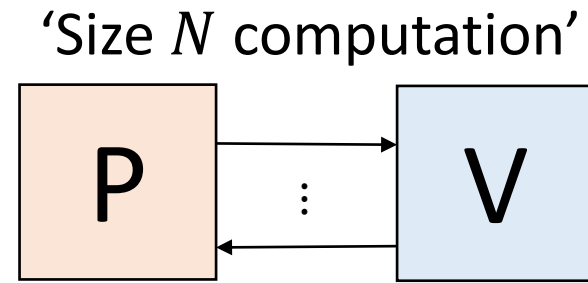
$V^*(G_0, G_1)$
???

But V^* may sample b from a strange distribution

Why is the simulator valid? (efficient, indistinguishable)

- S is efficient because V^* is, and we expect to clear Step 5 in 2 tries.
- $\tau(G_0)$ and $\tau(G_1)$ identically distributed as $G_0 \cong G_1$.
- Simulated τ, b are distributed as in a real protocol.
- $H = \tau(G_b)$ so H is uniquely determined and correctly distributed too.

Efficiency targets



1. Introduction and definitions	NP proofs	$\text{poly}(N)$ bits $\leftrightarrow$	$\text{poly}(N)$ V-time	Not ZK
2. Sigma protocols Efficient for small computations		$\text{poly}(N)$	$\text{poly}(N)$	ZK
3. ZK arguments with short proofs Standard assumptions		$\text{polylog}(N)$	$\text{polylog}(N)$ Main targets	ZK
4. Non-interactive zero-knowledge Strong assumptions		$O(1)$	$O(1)$	ZK
5. Bonus material?				

Life lessons

- Interaction is more powerful. You can achieve more by asking questions and having discussions.
- Don't have conversations that you can already simulate in your head – you won't learn anything!

Next time:

- Variations of soundness and zero-knowledge.
- Zero-knowledge for an NP-complete problem.